

WARNING!

This documentation contains cyber attack simulations conducted within a **controlled laboratory environment** against a **legitimate/authorized target**.

Performing cyber attacks in a **production (real-world) environment** or against **unauthorized targets** is illegal and **violates the law**.

The author **assumes no responsibility** for any misuse of the information contained within this documentation for illegal activities.

Table of Contents

Table of Contents	2
Attacker Profile	3
Target Profile.....	4
Profil Laboratorium.....	5
Fase 1: Reconnaissance	6
1.1. Target Discovery	6
Command: ip a.....	6
Command: netdiscover (sudo).....	6
Command: arp-scan (sudo).....	7
Command: ping.....	7
1.2. Port and Service Mapping.....	8
Command: nmap.....	8
1.3. Web Vulnerability Discovery	10
Command: nikto.....	10
1.4. Discovering Hidden Files via Web URL	11
Command: dirb (DirBuster)	11
APPENDIX.....	14

Attacker Profile

Parameter	Detail
Operating System Name	Kali Linux
Operating System Family	Linux
Operating System Version	2026.2
Base Operating System	Debian
Base Operating System Version	Debian 10.x 64-bit
Operating System Deployment	Virtual Machine (VM)
Virtualization Platform	VMware Workstation Pro

Target Profile

Parameter	Detail
Operating System Name	Metasploitable 2
Operating System Family	Linux
Operating System Version	2.6.X
Base Operating System	Ubuntu
Base Operating System Version	Ubuntu 8.04 LTS
Operating System Deployment	Virtual Machine (VM)
Virtualization Platform	VMware Workstation Pro

Profil Laboratorium

Parameter	Detail
Lab Name	VMware Workstation Pro

Fase 1: Reconnaissance

Reconnaissance (recon/information gathering) is the first phase in the cyber attack chain (*Cyber Attack Chain / Cyber Kill Chain*). In this phase, the attacker identifies the target, verifies that the target is active (alive/functioning), maps open ports (entry points), determines active services running on the target, and scans for vulnerabilities. The data collected during this phase will be utilized to design payloads/attacks (*weaponization*) and exploit target vulnerabilities in the subsequent phase.

1.1. Target Discovery

Command: ip a

`ip a` is short for `ip addr` or `ip address`. This command is used to display and configure IP addresses and network interfaces on the operating system where it is executed. In this documentation, `ip a` is used to quickly identify the target's IP address to expedite the laboratory process. In a real-world scenario, target administrators may use this command, but internal IP address information is generally confidential and not publicly shared. This command is executed on both the attacker and target machines for verification.

- Kali Linux IP address: 192.168.100.131/24
- Metasploitable 2 IP address: 192.168.100.132/24

Command: netdiscover (sudo)

`netdiscover` is an active/passive network scanning tool used to discover active hosts in a specific subnet. This command utilizes the ARP (**Address Resolution Protocol**) to identify devices by gathering details such as IP address, MAC address, packet count, and MAC vendor/hostname. This command is executed from the Kali Linux machine to detect the presence of the Metasploitable 2 machine.

- Network segment: 192.168.100.x/24
- IP address 192.168.100.1 belongs to the host OS running VMware.
- IP address 192.168.100.2 belongs to the VMware virtual gateway/router.
- IP address 192.168.100.132 belongs to the target machine (Metasploitable 2).
- IP address 192.168.100.254 belongs to the VMware internal DHCP Server.
- IP address 192.168.100.255 is the network broadcast address.

Command: arp-scan (sudo)

arp-scan is a command-line tool that systematically sends ARP Request packets to every IP address in the specified network segment. Similar to netdiscover, this tool maps IP addresses to MAC addresses in the local network to locate active hosts quickly.

- Network Segment: 192.168.100.x/24
 Interface: eth0 (ethernet0)
 Type: EN10MB
- IP Address 192.168.100.1 belongs to the host OS running VMware.
- IP Address 192.168.100.2 belongs to the VMware virtual gateway/router.
- IP Address 192.168.100.132 belongs to the target machine (Metasploitable 2).
- IP Address 192.168.100.254 belongs to the VMware internal DHCP Server.
- IP Address 192.168.100.255 is the network broadcast address.

Command: ping

The ping utility is used to verify whether a destination host is active and reachable over the network. It operates by sending ICMP (*Internet Control Message Protocol*) *Echo Request* packets to the target IP address and waiting for an *Echo Reply*. If the target responds, it indicates the connection is successful. If no reply is received (*Request Timed Out* or *Destination Host Unreachable*), the target is either offline, blocking ICMP packets (e.g., due to firewall rules), or experiencing a routing failure.

1.2. Port and Service Mapping

Command: nmap

nmap (*Network Mapper*) is a network security scanner used for network discovery and vulnerability auditing. The utility operates by scanning ports on the target device. A total of 65,535 ports are available on any network interface, categorized into three ranges:

1. **Well-Known Ports (0 - 1023):** Reserved for standard network services and core system protocols (e.g., HTTP, FTP, SSH).
2. **Registered Ports (1024 - 49,151):** Allocated for specific application and software vendors.
3. **Dynamic / Private Ports (49,152 - 65,535):** Dynamically allocated for temporary client connections.

In addition to detecting port status (open, closed, or filtered), nmap can perform service/version detection and target operating system detection.

Below is the list of open ports mapped by nmap:

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4

PORT	STATE	SERVICE	VERSION
22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
53/tcp	open	domain	ISC BIND 9.4.2
80/tcp	open	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	open	rpcbind	2 (RPC #100000)
139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp	open	exec	Netkit-rsh rexecd
513/tcp	open	login	OpenBSD or Solaris rlogind
514/tcp	open	tcpwrapped	-
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	Metasploitable root shell
2049/tcp	open	nfs	2-4 (RPC #100003)
2121/tcp	open	ftp	ProFTPD 1.3.1
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
3632/tcp	open	distccd	distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	open	vnc	VNC (protocol 3.3)

PORT	STATE	SERVICE	VERSION
6000/tcp	open	X11	(Access Denied)
6667/tcp	open	irc	UnrealIRCd
6697/tcp	open	irc	UnrealIRCd
8009/tcp	open	ajp13	Apache Jserv (Protocol v1.3)
8180/tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1
8787/tcp	open	drb	Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
38057/tcp	open	status	1 (RPC #100024)
42189/tcp	open	nlockmgr	1-4 (RPC #100021)
57819/tcp	open	mountd	1-3 (RPC #100005)
60910/tcp	open	java-rmi	GNU Classpath grmiregistry

Table 1.2.1 List of open ports from Nmap scan

1.3. Web Vulnerability Discovery

Command: nikto

nikto is an open-source web server vulnerability scanner designed to detect insecure configurations, default or sensitive files, and common vulnerabilities on web services (running on HTTP port 80 or HTTPS port 443). This scan is executed from the Kali Linux machine against the target's HTTP service.

- **Summary of Web Vulnerability Findings**

Based on the scan output, the following critical findings and vulnerabilities were identified on the target web server:

No	Finding Category	Finding Detail	Potential Risk / Impact
1	Outdated Server Software	Running Apache/2.2.8 and PHP/5.2.4 (outdated versions)	Vulnerable to exploitation of known public vulnerabilities
2	HTTP Methods Allowed	HTTP TRACE method is enabled on the server	Vulnerable to Cross-Site Tracing (XST) attacks to steal session cookies
3	Vulnerable Directories	/dav/ (WebDAV) directory is active with a weak configuration	Allows attackers to upload malicious files (web shells) to the server via the HTTP PUT method
4	Exposed Interfaces	Discovered login pages for phpMyAdmin, wiki (/twiki/), and other administrative utilities	Vulnerable to brute-force attacks on admin credentials or direct exploitation of web applications

Table 1.3.1 Summary of web vulnerability scan findings from Nikto

1.4. Discovering Hidden Files via Web URL

Command: dirb (DirBuster)

dirb (Directory Buster) is a command-line utility used to search for hidden directories and files on a website. It works by performing a dictionary-based brute-force attack using a predefined wordlist. If a directory or file in the list is found on the target web server, dirb reports the corresponding HTTP response code (such as 200 OK or 403 Forbidden). Directories configured with directory listing (allowing contents to be

viewed directly) will also be detected. This scan is run from the Kali Linux machine.

- **Summary of Critical Directories Found**

From the DIRB scan results, the following key directories were identified, serving as primary entry points for future exploitation:

Direktory	HTTP Code	Description / Function	Exploitation Significance
/dvwa/	302 / 200	Damn Vulnerable Web Application	Vulnerable web application for penetration testing practice; contains SQL injection, XSS, file inclusion, etc
/mutillidae/	302 / 200	OWASP Mutillidae II	Vulnerable web application designed specifically for OWASP Top 10 security practice
/phpMyAdmin/	302 / 200	MySQL administration interface	Provides database access if credentials are weak or if vulnerabilities in phpMyAdmin are exploited
/twiki/	200	Twiki collaboration platform	Outdated wiki platform vulnerable to Remote Code Execution (RCE)
/dav/	200	WebDAV directory	Writable directory that can be used to upload and execute payloads since write/PUT methods are enabled

Directory	HTTP Code	Description / Function	Exploitation Significance
/security /	200	Metasploitable security settings	Contains configuration info regarding the target site's security level

Table 1.4.1 Summary of critical directories found by DIRB

APPENDIX

```
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:fb:f8:f9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.131/24 brd 192.168.100.255 scope global dynamic noprefixroute eth0
        valid_lft 1315sec preferred_lft 1315sec
    inet6 fe80::4e73:669a:ff63:f2e5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 1: ip a command output on Kali Linux

```
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:ec:92:23 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.132/24 brd 192.168.100.255 scope global eth0
    inet6 fe80::20c:29ff:feec:9223/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop qlen 1000
    link/ether 00:0c:29:ec:92:2d brd ff:ff:ff:ff:ff:ff
```

Figure 2: ip a command output on Metasploitable 2

Currently scanning: Finished!		Screen View: Unique Hosts		
4 Captured ARP Req/Rep packets, from 4 hosts.		Total size: 240		
IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.100.1	00:50:56:c0:00:08	1	60	VMware, Inc.
192.168.100.2	00:50:56:e4:f8:8d	1	60	VMware, Inc.
192.168.100.132	00:0c:29:ec:92:23	1	60	VMware, Inc.
192.168.100.254	00:50:56:ed:10:09	1	60	VMware, Inc.

Figure 3: netdiscover command output on Kali Linux

```
(kali㉿kali)-[~]
$ sudo arp-scan --localnet
Interface: eth0, type: EN10MB, MAC: 00:0c:29:fb:f8:f9, IPv4: 192.168.100.131
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.100.1 00:50:56:c0:00:08 (Unknown)
192.168.100.2 00:50:56:e4:f8:8d (Unknown)
192.168.100.132 00:0c:29:ec:92:23 (Unknown)
192.168.100.254 00:50:56:ed:10:09 (Unknown)

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.876 seconds (136.46 hosts/sec). 4 responded
```

Figure 4: arp-scan command output on Kali Linux

```

(kali㉿kali)-[~]
$ ping 192.168.100.132
PING 192.168.100.132 (192.168.100.132) 56(84) bytes of data.
 64 bytes from 192.168.100.132: icmp_seq=1 ttl=64 time=1.80 ms
 64 bytes from 192.168.100.132: icmp_seq=2 ttl=64 time=0.698 ms
 64 bytes from 192.168.100.132: icmp_seq=3 ttl=64 time=0.636 ms
 64 bytes from 192.168.100.132: icmp_seq=4 ttl=64 time=0.724 ms
 64 bytes from 192.168.100.132: icmp_seq=5 ttl=64 time=0.842 ms
^C
— 192.168.100.132 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4066ms
rtt min/avg/max/mdev = 0.636/0.940/1.804/0.436 ms

```

Figure 5: Kali Linux pinging Metasploitable 2

```

(kali㉿kali)-[~]
$ nmap 192.168.100.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-25 09:31 -0400
Nmap scan report for 192.168.100.132
Host is up (0.0027s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:EC:92:23 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds

```

Figure 6: Kali Linux running nmap scan against Metasploitable 2 (Part 1)

```

(kali@kali)~$
$ nmap -A -p- 192.168.100.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-25 09:33 -0400
Nmap scan report for 192.168.100.132
Host is up (0.0014s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.100.131
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-date: 2026-07-25T13:35:49+00:00; +1s from scanner time.
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_SSL2_RC2_128_CBC_WITH_MD5
|_SSL2_DES_64_CBC_WITH_MD5
|_SSL2_DES_192_EDE3_CBC_WITH_MD5
|_SSL2_RC4_128_EXPORT40_WITH_MD5
|_SSL2_RC4_128_WITH_MD5
|_SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN

```

Figure 7: Kali Linux running nmap scan against Metasploitable 2 (Part 2)

```

53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_program version port/proto service
|_100000 2 111/tcp rpcbind
|_100000 2 111/udp rpcbind
|_100003 2,3,4 2049/tcp nfs
|_100003 2,3,4 2049/udp nfs
|_100005 1,2,3 38140/udp mountd
|_100005 1,2,3 57819/tcp mountd
|_100021 1,3,4 42189/tcp nlockmgr
|_100021 1,3,4 58652/udp nlockmgr
|_100024 1 38057/tcp status
|_100024 1 48809/udp status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
|_mysql-info:
|_Protocol: 10
|_Version: 5.0.51a-3ubuntu5
|_Thread ID: 8
|_Capabilities flags: 43564
|_Some Capabilities: ConnectWithDatabase, Support41Auth, SupportsTransactions, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, SupportsCompression, LongColumnFlag
|_Status: Autocommit
|_Salt: MH`L`Bgb4|9AzdfRGP
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))

```

Figure 8: Kali Linux running nmap scan against Metasploitable 2 (Part 3)


```

5432/tcp open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-07-25T13:35:49+00:00; +1s from scanner time.
5900/tcp open  vnc VNC (protocol 3.3)
|_vnc-info:
|_Protocol version: 3.3
|_Security types:
|_VNC Authentication (2)
6000/tcp open  X11 (access denied)
6667/tcp open  irc UnrealIRCd
|_irc-info:
|_users: 1
|_servers: 1
|_lusers: 1
|_lservers: 0
|_server: irc.Metasploitable.LAN
|_version: Unreal3.2.8.1. irc.Metasploitable.LAN
|_uptime: 0 days, 0:22:15
|_source ident: nmap
|_source host: 5F1669D2.2ADA0512.FFFA6D49.IP
|_error: Closing Link: jsumezyoy[192.168.100.131] (Quit: jsumezyoy)
6697/tcp open  irc UnrealIRCd
8009/tcp open  ajp13 Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
8787/tcp open  drb Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbb)
38057/tcp open status 1 (RPC #100024)
42189/tcp open nlockmgr 1-4 (RPC #100021)
57819/tcp open mountd 1-3 (RPC #100005)
60910/tcp open java-rmi GNU Classpath grmiregistry
MAC Address: 00:0C:29:EC:92:23 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

```

Figure 9: Kali Linux running nmap scan against Metasploitable 2 (Part 4)

```

Host script results:
|_ smb-os-discovery:
|_ OS: Unix (Samba 3.0.20-Debian)
|_ Computer name: metasploitable
|_ NetBIOS computer name:
|_ Domain name: localdomain
|_ FQDN: metasploitable.localdomain
|_ System time: 2026-07-25T09:35:41-04:00
|_ smb-security-mode:
|_ account_used: guest
|_ authentication_level: user
|_ challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ clock-skew: mean: 1h00m01s, deviation: 2h00m00s, median: 0s

TRACEROUTE
HOP RTT ADDRESS
1 1.43 ms 192.168.100.132

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 144.61 seconds

```

Figure 10: Kali Linux running nmap scan against Metasploitable 2 (Part 5)

```
(kali@kali)-[~]
$ nikto -h http://192.168.100.132
- Nikto v2.6.0

+ Target IP: 192.168.100.132
+ Target Hostname: 192.168.100.132
+ Target Port: 80
+ Platform: Unknown
+ Start Time: 2026-07-25 09:42:17 (GMT-4)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ ERROR: Failed to check for updates: 403
+ [999986] /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ [750500] /icons/: Directory indexing found.
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [999100] /index: Uncommon header(s) 'tcn' found, with contents: list.
+ [999965] /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives fo
it/sectou.php?id=4698ebdc59d15,https://exchange.xforce.ibmcloud.com/vulnerabilities/8275
+ [600625] PHP/5.2.4-2ubuntu5.10 appears to be outdated (current is at least 8.5.1).
+ [600050] Apache/2.2.8 appears to be outdated (current is at least 2.4.66).
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Optio
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [999967] /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ [000434] /: HTTP TRACE method is active and replies which suggests the host is vulnerable to XST. See: https://owasp.org/www-community/attacks/Cross_Site_
+ [750510] /phpinfo.php: Output from the phpinfo() function was found.
+ [750500] /doc/: Directory indexing found.
+ [001213] /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0678
+ [001384] /z-PHPB885F2A0-3C92-11d0-A3A9-4C7B08C10000: PHP Easter Eggs reveals potentially sensitive information via HTTP requests that contain specific QUE
-you-dare-to-show-your-php-easter-egg/
+ [001385] /z-PHPE9568F36-D428-11d2-A769-00AA001ACF42: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUER
you-dare-to-show-your-php-easter-egg/
+ [001386] /z-PHPE9568F34-D428-11d2-A769-00AA001ACF42: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUER
you-dare-to-show-your-php-easter-egg/
+ [001387] /z-PHPE9568F35-D428-11d2-A769-00AA001ACF42: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUER
you-dare-to-show-your-php-easter-egg/
+ [001795] /phpMyAdmin/changelog.php: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ [999984] /phpMyAdmin/changelog: Server may leak inodes via ETags, header found with file /phpMyAdmin/Changelog, inode: 92462, size: 40540, mtime: Tue Dec
ename.cgi?name=CVE-2003-1418
+ [001796] /phpMyAdmin/Changelog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
```

Figure 11: Kali Linux running nikto scan against Metasploitable 2 HTTP service
(Part 1)

```
+ [001796] /phpMyAdmin/Changelog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ [750500] /test/: Directory indexing found.
+ [001894] /test/: This might be interesting.
+ [002989] /phpinfo.php: PHP is installed, and a test script which runs phpinfo() was found. This gives a lot of system information. See: CWE-552
+ [003584] /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-icons/readme/
```

Figure 12: Kali Linux running nikto scan against Metasploitable 2 HTTP service
(Part 2)

```
(kali@kali)-[~]
$ dirb http://192.168.100.132

DIRB v2.22
By The Dark Raver

START_TIME: Sat Jul 25 09:45:13 2026
URL_BASE: http://192.168.100.132/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

GENERATED WORDS: 4612

--- Scanning URL: http://192.168.100.132/ ---
+ http://192.168.100.132/cgi-bin/ (CODE:403|SIZE:296)
=> DIRECTORY: http://192.168.100.132/dav/
+ http://192.168.100.132/index (CODE:200|SIZE:891)
+ http://192.168.100.132/index.php (CODE:200|SIZE:891)
+ http://192.168.100.132/phpinfo (CODE:200|SIZE:48107)
+ http://192.168.100.132/phpinfo.php (CODE:200|SIZE:48119)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/
+ http://192.168.100.132/server-status (CODE:403|SIZE:301)
=> DIRECTORY: http://192.168.100.132/test/
=> DIRECTORY: http://192.168.100.132/twiki/

--- Entering directory: http://192.168.100.132/dav/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/ ---
+ http://192.168.100.132/phpMyAdmin/calendar (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/changelog (CODE:200|SIZE:74593)
+ http://192.168.100.132/phpMyAdmin/Changelog (CODE:200|SIZE:40540)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/contrib/
+ http://192.168.100.132/phpMyAdmin/docs (CODE:200|SIZE:4583)
+ http://192.168.100.132/phpMyAdmin/error (CODE:200|SIZE:1063)
+ http://192.168.100.132/phpMyAdmin/export (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/favicon.ico (CODE:200|SIZE:18902)
+ http://192.168.100.132/phpMyAdmin/import (CODE:200|SIZE:4145)
```

Figure 13: Kali Linux running dirb scan against Metasploitable 2 HTTP service
(Part 1)

```

+ http://192.168.100.132/phpMyAdmin/changelog (CODE:200|SIZE:74593)
+ http://192.168.100.132/phpMyAdmin/ChangeLog (CODE:200|SIZE:40540)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/contrib/
+ http://192.168.100.132/phpMyAdmin/docs (CODE:200|SIZE:4583)
+ http://192.168.100.132/phpMyAdmin/error (CODE:200|SIZE:1063)
+ http://192.168.100.132/phpMyAdmin/export (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/favicon.ico (CODE:200|SIZE:18902)
+ http://192.168.100.132/phpMyAdmin/import (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/index (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/index.php (CODE:200|SIZE:4145)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/js/
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/lang/
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/libraries/
+ http://192.168.100.132/phpMyAdmin/license (CODE:200|SIZE:18011)
+ http://192.168.100.132/phpMyAdmin/LICENSE (CODE:200|SIZE:18011)
+ http://192.168.100.132/phpMyAdmin/main (CODE:200|SIZE:4227)
+ http://192.168.100.132/phpMyAdmin/navigation (CODE:200|SIZE:4145)
+ http://192.168.100.132/phpMyAdmin/phpinfo (CODE:200|SIZE:0)
+ http://192.168.100.132/phpMyAdmin/phpinfo.php (CODE:200|SIZE:0)
+ http://192.168.100.132/phpMyAdmin/phpmyadmin (CODE:200|SIZE:21389)
+ http://192.168.100.132/phpMyAdmin/print (CODE:200|SIZE:1063)
+ http://192.168.100.132/phpMyAdmin/readme (CODE:200|SIZE:2624)
+ http://192.168.100.132/phpMyAdmin/README (CODE:200|SIZE:2624)
+ http://192.168.100.132/phpMyAdmin/robots (CODE:200|SIZE:26)
+ http://192.168.100.132/phpMyAdmin/robots.txt (CODE:200|SIZE:26)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/scripts/
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/setup/
+ http://192.168.100.132/phpMyAdmin/sql (CODE:200|SIZE:4145)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/test/
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/themes/
+ http://192.168.100.132/phpMyAdmin/TOD0 (CODE:200|SIZE:235)
+ http://192.168.100.132/phpMyAdmin/webapp (CODE:200|SIZE:6903)

--- Entering directory: http://192.168.100.132/test/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

```

Figure 14: Kali Linux running dirb scan against Metasploitable 2 HTTP service
(Part 2)

```

--- Entering directory: http://192.168.100.132/twiki/ ---
=> DIRECTORY: http://192.168.100.132/twiki/bin/
+ http://192.168.100.132/twiki/data (CODE:403|SIZE:298)
+ http://192.168.100.132/twiki/index (CODE:200|SIZE:782)
+ http://192.168.100.132/twiki/index.html (CODE:200|SIZE:782)
=> DIRECTORY: http://192.168.100.132/twiki/lib/
+ http://192.168.100.132/twiki/license (CODE:200|SIZE:19440)
=> DIRECTORY: http://192.168.100.132/twiki/pub/
+ http://192.168.100.132/twiki/readme (CODE:200|SIZE:4334)
+ http://192.168.100.132/twiki/templates (CODE:403|SIZE:303)

--- Entering directory: http://192.168.100.132/phpMyAdmin/contrib/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/js/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/lang/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/libraries/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/scripts/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

--- Entering directory: http://192.168.100.132/phpMyAdmin/setup/ ---
+ http://192.168.100.132/phpMyAdmin/setup/config (CODE:303|SIZE:1370)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/setup/frames/
+ http://192.168.100.132/phpMyAdmin/setup/index (CODE:200|SIZE:8619)
+ http://192.168.100.132/phpMyAdmin/setup/index.php (CODE:200|SIZE:8627)
=> DIRECTORY: http://192.168.100.132/phpMyAdmin/setup/lib/
+ http://192.168.100.132/phpMyAdmin/setup/scripts (CODE:200|SIZE:21967)
+ http://192.168.100.132/phpMyAdmin/setup/styles (CODE:200|SIZE:6218)

--- Entering directory: http://192.168.100.132/phpMyAdmin/test/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

```

Figure 15: Kali Linux running dirb scan against Metasploitable 2 HTTP service
(Part 3)

```
— Entering directory: http://192.168.100.132/phpMyAdmin/themes/ —  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
— Entering directory: http://192.168.100.132/twiki/bin/ —  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
— Entering directory: http://192.168.100.132/twiki/lib/ —  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
— Entering directory: http://192.168.100.132/twiki/pub/ —  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
— Entering directory: http://192.168.100.132/phpMyAdmin/setup/frames/ —  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
— Entering directory: http://192.168.100.132/phpMyAdmin/setup/lib/ —  
(!) WARNING: Directory IS LISTABLE. No need to scan it.  
(Use mode '-w' if you want to scan it anyway)  
  
—  
END_TIME: Sat Jul 25 09:45:26 2026  
DOWNLOADED: 18448 - FOUND: 42
```

*Figure 16: Kali Linux running dirb scan against Metasploitable 2 HTTP service
(Part 4)*